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Experiment No 13

06/10

Title :- Data Visualization

Aim :- To apply and understand data visualization techniques using Matplotlib and Seaborn in Python.

Theory :-

Q1 What is the importance of data visualization in data science and machine learning?

=> Data visualization means presenting data in graphical form, such as charts and plots, so that patterns can be understood quickly. In data science it is used during exploratory analysis to find trends, relationships, skewness and outliers that are hard to notice in raw tables. It helps in checking data quality before cleaning and modelling, and in choosing suitable features and algorithms. After a model is trained, plots of accuracy, loss and predictions show how well it performs. Graphs also communicate the findings clearly to people who are not familiar with the technical details.

code :-

```
import matplotlib.pyplot as plt
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [20, 25, 31, 38, 44, 52]
plt.plot(months, sales, marker="o")
plt.xlabel("Month")
plt.ylabel("sales in thousands")
plt.title("Monthly Sales Trend")
plt.show()
```

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Q2 Differentiate between Matplotlib and seaborn libraries in Python.

⇒

Point	Matplotlib	seaborn
Nature	Basic, low level plotting library	High level library built on top of Matplotlib
code length	More lines are needed for a good looking plot	Fewer lines are needed.
Default look	Plain	Attractive themes and colour palettes.
Data input	lists and arrays	Works directly with pandas DataFrames using column names.
statistical plots	Have to be built Manually.	Ready functions such as boxplot, heatmap and pairplot.
control	Complete control over every Element	Less control, Matplotlib is used to fine tune.

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code :

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]
plt.bar(x, y, color = "steelblue")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Matplotlib Bar Plot")
plt.show()

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt.

df = pd.DataFrame({"Branch": ["AI", "IT", "CS", "EXTC"],
                   "Students": [60, 55, 50, 40]})
sns.barplot(x = "Branch", y = "students", data = df)
plt.title("Seaborn Bar plot")
plt.show()
```

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Q3 Explain the role of the following types of plots and mention one use case for each:

a) Bar chart
b) Histogram
c) Pie chart
d) Scatter plot

⇒ a) Bar chart:

A Bar chart uses rectangular bars whose heights show the value of each category, so categories can be compared easily. It is used to, for example, to compare the number of students in each branch.

b) Histogram:

A Histogram divides numeric data into bins and shows how many values fall in each bin, so it shows the shape of the distribution. It is used for example, to see how the marks of a class are spread.

c) Pie chart:

A pie chart divides a circle into slices whose sizes show the share of each category in the whole. It is used, for example, to show the percentage of a monthly budget spent on each head.

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d) Scatter Plot :

A Scatter Plot places each observation as a point using two numeric variables, so it shows the relationship between them. It is used, for example, to study how study hours are related to marks.

code :

```
import matplotlib.pyplot as plt

fig, ax = plt.subplot(2,2, figsize=(9,7), layout="constrained")
ax[0,0].bar(["AJ", "IT", "CS", "EXTC"], [60,55,50,40])
ax[0,0].set(title="Bar chart")

marks = [35,42,48,50,52,55,58,60,61,63,65,69,70,72,78,85]
ax[0,1].hist(marks, bin=5, edgecolor="black")
ax[0,1].set(title="Histogram")

ax[1,0].pie([40,25,20,15], autopct="%1.0f%%",
            labels=["Rent", "Food", "Travel", "Other"])
ax[1,0].set(title="Pie chart")

hours = [1,2,3,4,5,6,7,8]
score = [35,42,50,55,63,70,78,85]
ax[1,1].scatter(hours, score)
ax[1,1].set(title="Scatter Plot")

plt.show()
```

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Q4 what is the difference between a box plot and a histogram in terms of data interpretation?

⇒

Point	Box Plot	Histogram
shows	Median, quartiles, spread and outliers	frequency of values in each bin
Outliers	shown directly as separate points beyond the whiskers	seen only as small isolated bars far from the rest.
shape	gives a limited idea of shape and cannot show two peaks	shows skewness and the number of peaks clearly
comparing groups	Many boxes can be placed side by side easily	Becomes crowded for many groups.
dependence on bins	Doesn't depend on bins	appearance changes with the number of bins.

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code:

```
import matplotlib.pyplot as plt

data = [12, 15, 14, 16, 18, 17, 15, 14, 16, 19, 13, 15, 45]
fig, ax = plt.subplots(1, 2, figsize=(4, 4), layout="constrained")
ax[0].boxplot(data)
ax[0].set(title="Box plot")
ax[1].hist(data, bins=6, edgecolor="black")
ax[1].set(title="Histogram")
plt.show()
```

Conclusion:

Data visualization was performed using Matplotlib and seaborn to present data in graphical form. Bar charts, histograms, pie charts, scatter plots and box plots were created and the information given by each was understood. The choice of plot depends on the type of data and on the question that has to be answered.

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